

An Explicit ZFC Cardinal Obstruction to Laugwitz's Complete-Norm Question

Automatic Math Discovery Smoke Test

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Source and Extracted Question

The source is Renan J. S. Isneri, Josias V. Baca, and Lucas M. Fernandes, *Norms and Non-Equivalence in Infinite-Dimensional Banach Spaces*, arXiv:2508.15686. In the introduction the authors ask:

Does there exist some cardinal $\kappa > \aleph_0$ for which no vector space with a Hamel basis of that cardinality can be endowed with a norm that turns it into a Banach space?

They attribute the question to Laugwitz [3] and state that no explicit example of such a cardinal is currently known.

Every cardinal number $\kappa \leq \mathfrak{c}$, where $\mathfrak{c}$ denotes the cardinality of the continuum, there exists a Banach space F with cardinality κ . In such a case, any Hamel basis of F would necessarily have cardinality κ , as established in Theorem 3.5 of [7]. This would imply, in particular, that for any vector space E with a Hamel basis with cardinality κ , it would be possible to construct a linear isomorphism between E and F , thus endowing E with a Banach space structure. However, this approach fails when $\kappa = \aleph_0$, the cardinality of $\mathbb{N}$, as previously observed. This leads to an intriguing question: *Does there exist some cardinal $\kappa > \aleph_0$ for which no vector space with a Hamel basis of that cardinality can be endowed with a norm that turns it into a Banach space?* To the best of the authors' knowledge, no explicit example of such a cardinal

Answer

The cardinal

$$\kappa = \aleph_\omega$$

is an explicit ZFC example. No real or complex vector space with Hamel basis of cardinality $\aleph_\omega$ admits a norm whose metric completion is already the given space.

Idea of the Proof

There are two independent obstructions to a cardinal being a Banach-space Hamel dimension.

First, a theorem of Halbeisen and Hungerbuehler, cited by the source paper as [2, Theorem 3.5], says that every Hamel basis of an infinite-dimensional real or complex Banach space X has cardinality $|X|$. Thus a Banach Hamel dimension cannot be below the continuum.

Second, if X has density character δ , then the metric structure forces

$$|X| = \delta^{\aleph_0}.$$

Consequently any infinite-dimensional Banach Hamel dimension κ must satisfy $\kappa^{\aleph_0} = \kappa$. But by Koenig's theorem,

$$\aleph_\omega^{\aleph_0} > \aleph_\omega.$$

This excludes $\aleph_\omega$ whenever the continuum is at most $\aleph_\omega$, and the first obstruction excludes it whenever the continuum is larger.

Cardinality Lemma for Banach Spaces

Lemma 1. *Let X be an infinite-dimensional Banach space over $\mathbb{R}$ or $\mathbb{C}$, and let $\delta = \text{dens}(X)$. Then*

$$|X| = \delta^{\aleph_0}.$$

Proof. Let $D \subset X$ be dense with $|D| = \delta$. Every $x \in X$ is the limit of some sequence from D , so

$$|X| \leq |D|^{\aleph_0} = \delta^{\aleph_0}.$$

For the reverse inequality we first produce a separated family in the unit sphere of size δ . By transfinite induction choose unit vectors $(x_\alpha)_{\alpha < \delta}$ so that

$$\text{dist}(x_\alpha, \overline{\text{span}}\{x_\beta : \beta < \alpha\}) > 1/2.$$

This is possible because the closed linear span of fewer than δ vectors has density $< \delta$, hence is not all of X ; Riesz's lemma then supplies the next vector. Thus

$$\|x_\alpha - x_\beta\| > 1/2 \quad (\alpha \neq \beta).$$

For each sequence $s = (\alpha_n)_{n \geq 1} \in \delta^{\mathbb{N}}$, define

$$\Phi(s) = \sum_{n=1}^{\infty} 6^{-n} x_{\alpha_n} \in X.$$

The series converges absolutely. If $s \neq t$, let n be the first index with $\alpha_n \neq \beta_n$. Then

$$\begin{aligned} \|\Phi(s) - \Phi(t)\| &\geq 6^{-n} \|x_{\alpha_n} - x_{\beta_n}\| - 2 \sum_{m>n} 6^{-m} \\ &> \frac{1}{2} 6^{-n} - \frac{2}{5} 6^{-n} > 0. \end{aligned}$$

So Φ is injective and $|X| \geq \delta^{\aleph_0}$. □

The Obstruction

Theorem 1. *Let E be a real or complex vector space with Hamel dimension $\aleph_\omega$. No norm on E makes E into a Banach space.*

Proof. Suppose that some norm makes E a Banach space X .

Let $\mathfrak{c} = 2^{\aleph_0}$. If $\mathfrak{c} > \aleph_\omega$, then the underlying set of a real or complex vector space with Hamel dimension $\aleph_\omega$ has cardinality $\mathfrak{c}$. But Halbeisen–Hungerbuehler’s theorem gives

$$\dim_{\mathbb{H}}(X) = |X| = \mathfrak{c},$$

contradicting $\dim_{\mathbb{H}}(X) = \aleph_\omega$.

It remains to consider the case $\mathfrak{c} \leq \aleph_\omega$. Then $|X| = \aleph_\omega$, and again Halbeisen–Hungerbuehler gives

$$\aleph_\omega = \dim_{\mathbb{H}}(X) = |X|.$$

By Lemma 1, for $\delta = \text{dens}(X)$ one has

$$\aleph_\omega = |X| = \delta^{\aleph_0}.$$

Therefore

$$\aleph_\omega^{\aleph_0} = (\delta^{\aleph_0})^{\aleph_0} = \delta^{\aleph_0} = \aleph_\omega.$$

This contradicts Koenig’s theorem, since $\text{cf}(\aleph_\omega) = \aleph_0$ and therefore

$$\aleph_\omega^{\aleph_0} > \aleph_\omega.$$

The contradiction proves the claim. □

Corollary 1. *Every singular cardinal κ of countable cofinality gives the same kind of obstruction, after the identical case split for the continuum: if $\kappa \geq \mathfrak{c}$, Koenig’s theorem gives $\kappa^{\aleph_0} > \kappa$, and if $\kappa < \mathfrak{c}$, the Halbeisen–Hungerbuehler cardinality equality already rules it out.*

Verification and Novelty Notes

The proof is set-theoretic and does not require numerical or symbolic computation. The local run indexes were searched for the arXiv id 2508.15686, the phrases `Laugwitz`, `Hamel bases` of `Banach spaces`, and `aleph_omega`, and no duplicate solution packet was found. A local source search found the source paper’s citation of Halbeisen–Hungerbuehler and an independent Banach-space paper using the standard consequence $\aleph_\omega^{\aleph_0} > \aleph_\omega$ of Koenig’s theorem, but no packet answering the present Laugwitz question.

Bounded web searches for the arXiv id, the title, and close phrases involving Laugwitz, Hamel bases, Banach spaces, and $\aleph_\omega$ found no later explicit answer. Because the proof uses classical ingredients, the mathematical claim should be checked by a human reviewer and may be reclassified as a literature-implied answer if an exact prior formulation is located.

References

References

- [1] W. Arendt and R. Nittka, *Equivalent complete norms and positivity*, Archiv der Mathematik **92** (2009), no. 5, 414–427.
- [2] L. Halbeisen and N. Hungerbuehler, *The cardinality of Hamel bases of Banach spaces*, East-West Journal of Mathematics **2** (2000), no. 2, 153–159.

- [3] D. Laugwitz, *Über vollständige Normtopologien in linearen Räumen*, Archiv der Mathematik **6** (1955), no. 2, 128–131.
- [4] R. J. S. Isneri, J. V. Baca, and L. M. Fernandes, *Norms and Non-Equivalence in Infinite-Dimensional Banach Spaces*, arXiv:2508.15686.